ANKIT KUMAR

B.Tech. - Computer Science and Business Systems - Oriental Institute of Science and Technology, Bhopal

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PROFESSIONAL SUMMARY

Dedicated professional transitioning into [new field], bringing transferable skills and a strong commitment to learning and growth.

EDUCATION

Oriental Group of Institutes 2027

B.Tech. - Computer Science and Business Systems - Oriental Institute of Science and Technology, Bhopal | CGPA: **7.66** / **10**

R K SITA +2 HIGH SCHOOL HARIHARGANJ, Hariharganj 2022

12th | JAC | Percentage: **82.20** / **100**

ST J PUB SCHL K N MARKET WARISALIGANJ NAWADA BR, Nawada 2020

10th | CBSE | Percentage: **67.20** / **100**

PUBLICATIONS / RESEARCH / WHITE PAPERS

Smart Tourist Safety Monitoring and Incident Response System 18 Apr, 2026

Research paper | IEEE CONFERENCE | **Mentor:** Prof. Rakesh jaiswal | **No. of Authors:** 4

Key Skills: React Native, TensorFlow Lite, PyTorch, Kafka, MQTT, Edge Computing, Blockchain, Geofencing, Real-time Processing

Tourism safety has become one of the highest priorities for smart cities, government authorities,

and tourism boards. Traditional safety mechanisms lack real-time tracking, intelligent incident detection, secure identity authentication, and automated alert systems. To address these challenges, this project proposes a Smart Tourist Safety Monitoring & Incident Response System integrating three cutting-edge technologies: Artificial Intelligence (AI), Geo-Fencing, and Blockchain-based Digital Identity Verification.

The system aims to ensure tourist safety by:

Monitoring movement through secure GPS tracking

Issuing instant alerts upon entering unsafe or restricted zones

Detecting abnormal or distressing behavior using AI models

Providing tamper-proof digital identity via Blockchain

Offering an emergency SOS mechanism that reduces response time

Displaying real-time insights and alerts on a unified monitoring dashboard

Experimental analysis shows that geo-fencing reduces unauthorized zone entry by 80%, while blockchain ensures 100% tamper-proof identity verification. AI-based behavioral models improve emergency detection accuracy and shorten response time from 15 minutes to nearly 5 minutes. The proposed solution demonstrates an innovative approach for building safe, trustworthy, and smart tourism infrastructure

Smart Waste Segregation and Recycling System Using IOT and Machine Learning 22 Mar, 2026

Research Paper | IEEE | **Mentor:** Pro. Rakesh Jaiswal | **No. of Authors:** 2

Key Skills: Machine Learning | C++ | Computer Networking | Javascript

Rapid urbanization has intensified the global waste management crisis, making efficient waste segregation at the source an urgent priority. This paper presents a Smart Waste Segregation and Recycling System (SWSRS) that integrates Internet of Things (IoT) sensors with machine learning (ML) algorithms to automatically classify household waste into three categories: organic, recyclable, and hazardous. The proposed system employs an ensemble model combining Convolutional Neural Networks (CNN) for image-based classification and Random Forest for sensor-based data fusion, achieving an overall classification accuracy of 97.8%. A real-time feedback mechanism notifies users and synchronizes with the municipal waste management infrastructure for optimized collection scheduling. Experimental validation conducted on a dataset of 12,000 waste samples demonstrates significant improvements over existing methods.

ACHIEVEMENTS

- 1st Runner-Up in badminton competition at state level tournament under 19 in 2017
- 1st prize in debate competetion at sate level in 2017

SEMINARS / TRAININGS / WORKSHOPS

Pre-placement Talk internship **Institute Name:** Oriental Institute of Science and Technology 23 Apr, 2026 - 23 Apr, 2026

Key Skills: Basic AI knowledge, ML fundamentals, NLP basics, data analysis, ethical AI awareness, problem solving, quick learning

I had the opportunity to attend an insightful pre-placement talk conducted by Tata Consultancy Services (TCS), where the primary focus was on the growing importance of Artificial Intelligence (AI) in the modern technological landscape. The session was led by Mr. Vipin Mahazan, a Senior Associate at TCS, who shared his valuable industry experience and knowledge regarding AI applications, trends, and career opportunities.

The seminar began with an introduction to Artificial Intelligence, where AI was explained as the simulation of human intelligence processes by machines, especially computer systems. Mr. Mahazan emphasized that AI is not just a futuristic concept but a present-day reality that is transforming industries across the globe. He highlighted how AI systems are capable of performing tasks such as learning, reasoning, problem-solving, perception, and language understanding.

One of the key areas discussed during the session was the classification of AI. He explained the difference between Narrow AI, General AI, and Super AI. Narrow AI refers to systems designed to perform specific tasks such as voice assistants and recommendation systems. General AI, although still under development, aims to perform any intellectual task that a human can do. Super AI is a hypothetical concept where machines surpass human intelligence in all aspects. This classification helped in understanding the current stage and future potential of AI.

The session further explored the major domains of AI, including Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), and Computer Vision. Machine Learning was explained as a subset of AI that allows machines to learn from data without being explicitly programmed. Deep Learning, a further subset of ML, uses neural networks with multiple layers to analyze complex patterns. NLP enables machines to understand and process human language, while Computer Vision allows systems to interpret visual data such as images and videos.

Mr. Mahazan provided real-world examples to illustrate the impact of AI. He discussed applications such as chatbots, recommendation engines (like those used by e-commerce platforms), fraud detection systems in banking, self-driving cars, and healthcare diagnostics. He also explained how AI is used in industries like finance, healthcare, retail, manufacturing, and education to improve efficiency, accuracy, and decision-making.

A significant portion of the talk was dedicated to the use of AI in TCS projects. He explained how TCS integrates AI into its solutions to help clients automate processes, enhance customer experience, and gain insights from large datasets. He mentioned the use of AI in predictive analytics, where companies can forecast trends and make data-driven decisions. Additionally, he highlighted AI-powered automation tools that reduce manual effort and increase productivity.

The speaker also discussed the importance of data in AI. He emphasized that data is the backbone of AI systems, and the quality of data directly impacts the performance of AI models. He explained the concepts of structured and unstructured data and how data preprocessing is essential before training machine learning models. This gave a clear understanding of how raw data is transformed into meaningful insights.

Another important topic covered was the lifecycle of an AI project. Mr. Mahazan explained the various stages, including problem identification, data collection, data preprocessing, model selection, training, evaluation, and deployment. He stressed the importance of continuous monitoring and improvement of AI models to ensure their effectiveness over time.

The seminar also addressed the ethical considerations of AI. Issues such as data privacy, bias in AI models, and the potential impact of automation on jobs were discussed. Mr. Mahazan emphasized the need for responsible AI development and the importance of creating transparent and fair AI systems. He encouraged students to consider ethical implications while working on AI projects.

In addition to technical knowledge, the session provided valuable career guidance. Mr. Mahazan shared insights into the skills required to build a career in AI and related fields. He emphasized the importance of programming languages such as Python, knowledge of data structures and algorithms, and familiarity with AI frameworks and tools. He also highlighted the importance of problem-solving skills, critical thinking, and continuous learning.

He advised students to work on real-world projects, participate in hackathons, and build a strong portfolio to enhance their employability. He also suggested exploring online courses and certifications in AI and Machine Learning to gain practical knowledge. Networking and staying updated with industry trends were also highlighted as essential factors for career growth.

The session concluded with a discussion on future trends in AI. Mr. Mahazan talked about advancements such as generative AI, AI in cybersecurity, edge AI, and the integration of AI with emerging technologies like IoT and blockchain. He explained that AI will continue to evolve and create new opportunities in various domains.

Overall, the pre-placement talk was highly informative and provided a comprehensive understanding of Artificial Intelligence and its real-world applications. It not only enhanced my technical knowledge but also gave me clarity on career opportunities in the field of AI. The insights shared by Mr. Vipin Mahazan were valuable in understanding industry expectations and preparing for a future in the IT sector. Attending this seminar has motivated me to explore AI further, develop relevant skills, and work on practical projects. It has also helped me understand the importance of innovation and adaptability in today's fast-changing technological environment.

Attenttending the IEEE Conference 18 Apr, 2026 - 18 Apr, 2026

Institute Name: Oriental Institute of Science and Technology

Key Skills:

IoT, ML, Data Analysis, Sensors, Problem Solving, Teamwork, Communication, Research, Presentation, Critical Thinking

Oriental Institute of Science and Technology (OIST)

Project Title: Smart Waste Segregation and Recycling System using IoT and Machine Learning

I actively participated in an IEEE Conference organized at Oriental Institute of Science and Technology, where I, along with my teammate, successfully presented a research-based project focused on smart waste management. This conference provided a valuable platform to showcase innovative ideas, enhance technical knowledge, and interact with professionals and peers in the field of emerging

technologies.
Our project, titled “Smart Waste Segregation and Recycling System using IoT and Machine Learning,” was designed to address one of the most pressing environmental issues—inefficient waste management. The primary objective of the project was to develop an intelligent system capable of automatically segregating waste into different categories such as biodegradable, non-biodegradable, and recyclable materials using advanced technologies.
During the development phase, we conducted extensive research on existing waste management systems and identified their limitations, including manual segregation, inefficiency, and lack of automation. Based on our findings, we proposed a smart solution that integrates Internet of Things (IoT) devices and Machine Learning algorithms to automate the segregation process and improve overall efficiency.
The system was designed using sensors, microcontrollers, and machine learning models. The IoT component enabled real-time monitoring and data collection, while the machine learning model was trained to classify waste materials based on their characteristics such as shape, size, and texture. This combination allowed the system to make intelligent decisions and perform automated segregation with minimal human intervention.
We also focused on designing a user-friendly interface and efficient workflow for the system. The collected data could be monitored remotely, allowing authorities to track waste levels and optimize collection processes. This approach not only improves waste segregation but also contributes to better recycling practices and environmental sustainability.
As part of the conference, we prepared a detailed PowerPoint presentation (PPT) to explain our project concept, methodology, implementation, and results. The presentation included diagrams, system architecture, workflow, and real-world applications. We ensured that our explanation was clear, structured, and technically sound to effectively communicate our ideas to the audience and evaluators.
During the presentation, we confidently demonstrated our project, explained the working model, and answered questions from the judges and participants. Our ability to clearly articulate the concept, justify our approach, and handle technical queries played a key role in our success.
Our project received an A++ grade, which reflects the quality of our research, innovation, and presentation skills. This achievement highlighted our strong teamwork, technical understanding, and problem-solving abilities.
Through this experience, I developed and enhanced several important skills, including:
Research and analytical thinking
Problem-solving and innovation
Team collaboration and coordination
Technical presentation and communication skills
Knowledge of IoT and Machine Learning concepts
Practical understanding of real-world applications of technology
The conference also provided an opportunity to learn from other participants, explore new ideas, and gain insights into current trends in technology and research. Interacting with other teams and experts helped broaden my perspective and motivated me to work on more innovative projects in the future.
This experience was a significant milestone in my academic journey, as it allowed me to apply theoretical knowledge to a practical problem and present it on a professional platform. It strengthened my confidence, improved my communication skills, and enhanced my ability to work effectively in a team environment.
Overall, participating in this IEEE conference was a highly enriching experience that contributed to both my personal and professional growth. It not only improved my technical skills but also prepared me for future challenges in the field of engineering and technology.

EXTRA CURRICULAR ACTIVITIES

- Participated in college/school cultural events like in coding competiton
- Contributed to organizing annual functions and events
- Took part in competitions like singing, debate, or art

PERSONAL INTERESTS / HOBBIES

- Playing batminton
- Listening to music Watching movies
- Traveling and exploring new places
- Reading books

PERSONAL DETAILS

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Marital Status: Single
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